Annotating samples

Telling the tool which samples belong together, which is what makes every later plot mean something.

AT A GLANCE

Where	Data → Annotation
Produces	The metadata table, the $\log_2$ matrix and the filtered matrix, for one level at a time.
Needed by	Conditions, colours, sample labels, filtering, imputation, every statistical test and every summary plot.
Two routes	Assign columns to conditions by hand, or upload a metadata file. An uploaded file wins.
Optional	No. Until a level is annotated it has no $\log_2$ matrix, and most pages have nothing to draw.
Pairing	Stored here, as pair_id; edited on the pages that run paired tests.

1 What the annotation is

The annotation is a table with one row per sample: the name of the dataset column, and the condition that column belongs to. Everything the tool knows about the experiment is in it. Generating it does three things at once — it writes that table, it builds the $\log_2$ matrix from the columns the table names, and it filters that matrix by the settings beside the button.

The metadata table

Column	What it holds
sample	The dataset column, exactly as it is spelled in the uploaded file. This is the *original sample name*.
condition	The condition the sample belongs to. Case and spelling are taken as given: Control and control are two conditions.
pair_id	One identifier shared by the two samples of a pair. Present only once a pairing has been saved, or once an uploaded file supplied the column.

A sample appears exactly once. Assigning one column to two conditions is refused by name, and so is a metadata file that lists a sample twice — identically, because a file that repeats a row was assembled wrong, and with different values, because the file then states two incompatible facts and row order would decide which one wins.

What the samples are called afterwards

Plots do not use the original sample name. Each sample is relabelled condition_n, where *n* is its 1-based position among the samples of its own condition in metadata order — so the fourth sample of WT is WT_4 whether it is the fourth or the ninth column of the file. Numbering restarts in every condition. The original name is carried alongside, and **Toggle IDs** on a plot page adds it under the label on a second line; the tables that list samples carry it as Original_Name.

2 Annotating by hand

1. Choose the level in **Dataset level**. Only levels that the active package actually holds are offered.
2. In **Manual Condition Mapping**, type a **Condition name** and select its **Sample columns** from the list. The list is every column of the uploaded table, so annotation columns are offered too and must simply not be selected.
3. **Add condition** for the next one. **Remove** deletes a condition block and its assignments.
4. Check **Dataset is already log2 transformed**. It is set for you as soon as the first samples are assigned, and is only guessed at until you touch it — see below.
5. Set the filter, and press **Generate Annotation**.

A name copied with a trailing space still matches, whitespace ignored. Anything that still does not match is dropped, and the warning band names it and states how many samples were annotated instead.

> **IF NOTHING IS ASSIGNED**
>
> Generating with no conditions is not an error if the column names carry dates. Of the columns containing an eight-digit string beginning 20, those whose name also says quantity, intensity, abundance, area or lfq become the samples when there are any; otherwise every dated column becomes a sample except the dated per-run statistics — q-values, p-values, counts, NrOf columns, identified, scores and PEP. All of them go into one condition called sample, and the warning band says the metadata was auto-generated. It exists so a new dataset can be looked at before it is described, not so a test can be run against it.

3 Annotating from a file

Metadata file takes the same five file types as a dataset. It needs a sample column — recognised as sample, sample_name, data_column_name, data column name, data_column, column_name or column, and otherwise taken to be the first column of the file — and at least one further column to group by. A column called condition is used as the condition if one is present; otherwise the first grouping column is.

Each grouping column becomes its own *metadata profile*, named after the column, and a file with several of them therefore describes the same samples several ways at once — by treatment, by timepoint, by replicate batch — with automatic combinations generated on top. At most five grouping columns are accepted, and at most five manual profiles exist at a time; the annotation manager shows the count.

While a metadata file is active for a level it supplies the conditions, and the manual mapping below is kept but not used. The page says so at the top of the mapping block.

Rows naming a sample the dataset does not have are dropped rather than refused — a shared metadata file legitimately covers more runs than are loaded — and the warning band names them, because three mistyped names silently turn a 6-vs-6 design into 6-vs-3. Rows with a blank sample or condition are dropped, a trailing .0 is stripped from a sample name because that is what a spreadsheet does to a numeric run identifier, and a file matching no column at all is refused.

> **WORTH KNOWING**
>
> Uploading a metadata file for the phospho level also annotates the phosphoprotein level, which shares it. There is no separate phosphoprotein metadata upload.

4 The log2 question

The tool has to know whether the uploaded quantities are already on the $\log_2$ scale, because it will otherwise take the logarithm of a logarithm. **Dataset is already log2 transformed** states the answer. Until it is clicked, it is answered by detection over the assigned sample columns only:

> **HOW THE SCALE IS DETECTED**
>
> Column names are weak evidence and are trusted only when *every* assigned column carries log2 as a token, and never over a clear contradiction from the values.
> The values are read next. If at least 0.1% of them are negative, the data are $\log_2$. If the 1st and 99th percentiles both fall in $[0, 2]$, no answer is given: that range is equally consistent with ratios and with transformed data. Otherwise each column votes —

linear if its 95th percentile exceeds 50 or its 99th exceeds 100, $\log_2$ if its median is at most 35, its 95th percentile at most 45 and its 99th at most 60 — and a majority of the voting columns decides.
When nothing decides, generating the annotation is refused with the reason, and the checkbox must be set by hand.

Whatever the outcome, it is stated in the warning band with the evidence behind it. Setting the checkbox yourself removes the guess, and the detection stops running for that dataset.

5 Pairing samples

A paired test needs to know which sample of one condition corresponds to which sample of the other — the same donor, the same passage, the same day. That correspondence is stored in the annotation as `pair_id`: two samples carry the same identifier when they are a pair.

It is edited in the **Pair samples** panel, which appears on the pages that can run a paired analysis — statistical analysis, the phospho-specific pages, Kinase Library and the analysis canvas — rather than here, because the two conditions being paired are chosen there. Each row of the panel is one sample of the first condition with a selector for its partner in the second, and each sample can be used once: a sample already chosen is disabled in every other row. **Leave unpaired (excluded)** drops a sample from the comparison, and the footer counts what has been excluded on each side. Two pairs are the minimum.

Save pairing writes the identifiers into the annotation, and into the uploaded metadata if there is one; until it is pressed the panel says so and the analysis will not run on the mapping. An identifier the file supplied is kept, and one the panel has to invent is called `ui-pair-1`, `ui-pair-2` and so on. Identifiers are shared across conditions, so with more than two conditions saving a mapping also fixes those samples' partners in later comparisons; the panel warns that it will.

Regenerating the annotation preserves the pairing, but only when the sample-to-condition map came back identical. Move one sample to another condition and the identifiers are dropped, because a pairing across a design that has changed is no longer a pairing.

6 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which level to annotate: Protein, Phospho, Phosphoprotein or Peptide. Only loaded levels are listed.	the first loaded level	page
Dataset is already log2 transformed	States the scale of the uploaded quantities. Set by detection until it is clicked.	on, then detected	page
Metadata file	A sample-to-condition table to upload. Same file types as a dataset.	none	page
Upload Metadata	Reads the file and splits its grouping columns into metadata profiles.	—	page
Condition name	The name of one condition. Used verbatim as the condition label everywhere.	empty	page
Sample columns	The dataset columns belonging to that condition. Multiple selection. Called **Peptide file names** at peptide level.	none	page
Add condition	Adds an empty condition block.	—	page
Remove	Deletes one condition block and its assignments.	—	page
Generate Annotation	Builds the metadata table, the $\log_2$ matrix and the filtered matrix. **Generate Peptide Annotation** at peptide level.	—	page
Filter method, Minimum valid values, Scope, Normalization	The completeness filter applied while generating. Described in *Filtering*. At phosphoprotein level no **Normalization** control is shown: phosphoprotein data is never normalized at that level.	see there	page
Show Annotation Manager	Reveals the four controls below. The choice is remembered between visits.	hidden	page
Selected metadata	The active metadata profile. Switching it switches the conditions every page sees, without re-uploading anything.	Metadata	manager
New metadata profile	Creates an empty profile. At most five manual profiles exist at once, and the count is shown under the field.	empty	manager
Rename selected metadata	Renames the active profile.	the active profile	manager
Delete	Deletes the active profile and every automatic combination built from it. Asks first.	—	manager

EASY TO GET WRONG

A condition has two names and they are not the same thing. The one typed here is the condition itself: it decides which samples are compared, which colour they share, what the statistics call the groups, and it changes every page at once. The one in the plot appearance panel under **Condition names** is a display override — it changes the text printed on one figure and nothing else. Rename here to change the experiment; rename there to fix a label for a poster.

7 What the page shows afterwards

Result Summary, the warning band, and previews of the filtered matrix and the metadata table with a **Download Full CSV** button each — all described in *What is in the dataset*. At phospho level, **Aggregate to Phosphoprotein** follows; see *Phosphosites to phosphoproteins*.

Behind this page

Function	What it produces
compute_annotation	The manual route: metadata, $\log_2$ matrix, filtered matrix
compute_annotation_from_metadata	The same, from an uploaded metadata table
build_metadata_from_conditions	Turns condition blocks into the metadata table
normalize_uploaded_metadata	Validates an uploaded metadata file against the loaded dataset
split_uploaded_metadata_profiles	One metadata profile per grouping column
detect_log2_transformed	Decides the scale of the assigned sample columns
sample_label_frame	Maps each original sample name to its condition_n label
update_annotation_pairing	Writes pair_id from the pairing panel
paired_sample_columns	Resolves the two matched sample orders for a paired test